Computer Vision - HW Assignment No. 1

Lazăr Ionuț-Radu

October 2026

1) Why do we prefer AI models that learn an inner model of the world and how it works (from a geometric, temporal, dynamic, physics, and semantics perspective)?

2) Why is it not enough to learn to predict the right class or action directly from patterns of data, without knowing 'why' or having a model of how the world 'works' or 'is'?

Answer: To answer these and the following questions, I should begin by explaining what I interpret "inner model of the world" to mean. Intuitively, I view neural networks as complex (continuous) transformations of the data space, usually followed by a final linear layer. For a prediction problem, this is equivalent to drawing hyperplanes in the latent space to separate classes, for a regression problem, this is linear regression in the latent space.

I interpret the "inner model of the world" to be related to this transformation to a latent space. The "simpler" the data distribution or data manifold gets when projected into latent space, the better its performance. Through this lens, any deep learning model would build an inner model of the world, that is shaped differently by whichever learning method you choose to apply.

Training a model to distinguish between images of cats and dogs will "learn" features that are less useful than features "learnt" by a model trained on ImageNet dataset. For example, the pretrained Inception-v4 [8] model has proven itself useful in image generation literature, where it is used as a feature extractor for the FID (Frechet Inception Distance) metric [4].

Often times, a different learning objective can translate to features that are useful for your problem. The DINO [1] self-supervised method takes an image, augments it in 2 different ways that are fed into a student and a teacher model. The student is trained to minimize the difference between the 2 latents (of the 2 augmented images), while the teacher is an EMA of the student. As their paper shows, it turns out that this method learns semantic segmentation without any supervision.

Moving away from this data modality, in natural language processing BERT [3] and GPT [7] provide useful pretraining objectives that are beneficial as a starting point for many applications. Encoding semantically similar phrases through them results in latents that are close to each other (where the metric is cosine similarity), being useful in RAG systems. This is similar to how word2vec encoding results in some nice "arithmetic" properties of latents [5]:

$$v_{\text{Queen}} \approx v_{\text{King}} + v_{\text{Woman}} - v_{\text{Man}}$$

So, the learning of an inner model of the world relates directly to the performance and range of applicability of the model. Different learning objectives that are not necessarily directly related to the problem may prove beneficial for more use cases, and may help improve performance.

3) What is the advantage of having a model that considers many tasks and has many different interpretations of the world at the same time (it estimates geometry, movement of objects, as well as, their classes, together) as opposed to completely different models for different tasks?

Answer: As seen before, the answer to a question may benefit another. There are features that may be useful for both (or all, if there are more than 2) tasks, and there may be features that are not best suited for this "multidisciplinary" task. Thinking in this loose, informal way, it would not only lead to better results using a shared latent space for all tasks, but it could also discover, in a more informed way, which features are useful across the board, perhaps leading to a model that is smaller than the several you would otherwise need.

While I do not have a perfect example for this argument, something that comes to mind is the CLIP model [6], where image-text pairs are projected using 2 encoders into a shared latent space, with the objective of minimizing the distance between corresponding pairs of latents and maximizing the distance between any others. Adding the denser information content of the text prompts can only help in giving the encoder a more general inner model of the data. This, in itself, can be used as a starting point for purely image-related tasks.

Somehow unrelated to the question, but this may also help bridge the gap between models using different data modalities; image generation from a text prompt [2]

4) Why do we prefer intelligence that learns and discovers knowledge by itself and not just supervised by a teacher or known ground truth during training?

Answer: Learning from a dataset of supervised by a teacher can be restricting, in the sense that there is an upper limit on performance defined by quality / size of the dataset or performance of the teacher model. That being said, I believe there is a right place for many approaches. Distilling knowledge from a teacher model may be preferred when there are memory or speed of inference constraints, having an as good as possible smaller and faster model. Otherwise, finding a best and most general method is, of course, preferred, if possible avoiding using hard to make datasets, which in many cases may be inherently hard to make (for example, medical datasets are imbalanced; pathological classes will naturally happen less often).

References

[1] Mathilde Caron, Hugo Touvron, Ishan Misra, Hervé Jégou, Julien Mairal, Piotr Bojanowski, and Armand Joulin. Emerging properties in self-supervised vision transformers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 9650–9660, 2021.

[2] Katherine Crowson, Stella Biderman, Daniel Kornis, Dashiell Stander, Eric Hallahan, Louis Castricato, and Edward Raff. Vqgan-clip: Open domain image generation and editing with natural language guidance, 2022.

[3] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of NAACL-HLT*, pages 4171–4186, 2019.

[4] Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. Gans trained by a two time-scale update rule converge to a local nash equilibrium. In *Advances in Neural Information Processing Systems*, volume 30, pages 6626–6637, 2017.

[5] Tomas Mikolov, Wen-tau Yih, and Geoffrey Zweig. Linguistic regularities in continuous space word representations. In *Proceedings of NAACL-HLT*, pages 746–751, 2013.

[6] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision, 2021.

[7] Alec Radford, Karthik Narasimhan, Tim Salimans, and Ilya Sutskever. Improving language understanding by generative pre-training. 2018.

[8] Christian Szegedy, Sergey Ioffe, Vincent Vanhoucke, and Alexander Alemi. Inception-v4, inception-resnet and the impact of residual connections on learning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 31, 2017.